# Discrete-Time Markov Chains (DTMCs): Part 2

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# References


- S.M. Ross, “Stochastic Processes”: Chapter 4
- R. Wolff: Chapter 3
- S.M. Ross, “Introduction to Probability Models”: Chapter 4
- M. Harchol-Balter: Chapters 8 and 9
- A. Kumar, “Discrete Event Stochastic Processes”: Chapter 2

# Recall: Probability Vectors and Transition Probability Matrices

- For $j \geq 0$:
  - Let $P(X_0 = j) = a_j$
  - For $n \geq 1$, let $P(X_n = j) = a_j^{(n)}$
- Consider the following row vectors:
  - $a = [a_0, a_1, \dots]$
  - $a^{(n)} = \left[a_0^{(n)}, a_1^{(n)}, \dots\right]$
- $a$ and $a^{(n)}$ are called *probability vectors* since their elements are non-negative and sum to $1$
- Suppose transition probability matrix is $P$
- $a^{(n)}$:
  - $a^{(n-1)}P$
  - $aP^n$

# Example

- Suppose $X_n$ is brand chosen on a consumer's $n$th purchase:
  - ❑ modeled as a DTMC
- Market share of brand $j$:
  - ❑ fraction of all purchases that are of brand $j$

$$1) \quad \lim_{n\to\infty} \frac{\sum_{s=1}^{n} 1_{\{X_s=j\}}}{n}$$

- Expectation of limit in 1):

$$2) \quad \lim_{n\to\infty} \frac{\sum_{s=1}^{n} a_j^{(s)}}{n}$$

- Alternative definition of market share:

$$3) \quad \lim_{n\to\infty} a_j^{(n)} = \pi_j, \text{ say,}$$

  - ❑ where $\pi_j \geq 0$, for all $j$, and $\sum_j \pi_j = 1$

- We have assumed that for each $j$, the limit in 3) exists and is independent of initial vector $a$
- Recall:

$$4) \quad a^{(n)} = a^{(n-1)}P$$

- Taking limits as $n \to \infty$ on both sides of 4), we get:

$$5) \quad \pi = \pi P$$

- Also, by 3), it follows that:
  - ❑ the limit in 2) equals $\pi_j$
- Thus, market share of brand $j$ can be found by solving the system of equations 5) along with $\sum_j \pi_j = 1$

# Example (contd.)

- Recall:

  *1)* $\lim_{n\to\infty} \frac{\sum_{s=1}^{n} 1_{\{X_s=j\}}}{n}$

  2) $\lim_{n\to\infty} \frac{\sum_{s=1}^{n} a_j^{(s)}}{n}$

  *3)* $\lim_{n\to\infty} a_j^{(n)} = \pi_j$

  *4)* $a^{(n)} = a^{(n-1)}P$

  *5)* $\pi = \pi P$

- It is possible that limit in 3) does not exist, but limit in 2) can be found by solving the system of equations 5)
- E.g., let:
  - ❑ $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, and
  - ❑ $a = [a_0, a_1]$, where $a_0 + a_1 = 1$
- $a^{(n)}$:
  - ❑ $[a_0, a_1]$ for $n$ even
  - ❑ $[a_1, a_0]$ for $n$ odd
- Limit in 2):

  *6)* $\pi_0 = \pi_1 = \frac{1}{2}$

- The vector in 6) is also the unique solution of 5) that is a probability vector
- Limit in 3):
  - ❑ does not exist, unless we choose $a = \left[\frac{1}{2}, \frac{1}{2}\right]$

# Positive Recurrence of Irreducible DTMCs

- Recall:
  - **Theorem**: If $i \leftrightarrow j$, then:

    *a)* $P\left\{\lim_{t\to\infty} \frac{N_j(t)}{t} = \frac{1}{\mu_{jj}} \middle| X_0 = i\right\} = 1$

    *b)* $\lim_{t\to\infty} \frac{M_j(t)}{t} = \lim_{t\to\infty} \frac{\sum_{n=1}^{t} P_{ij}^n}{t} = \frac{1}{\mu_{jj}}$
- **Theorem**:
  - For an irreducible DTMC, all states are positive recurrent iff there is a unique stationary probability vector $\pi$ satisfying $\pi = \pi P$:
    - where $\pi_j = \frac{1}{\mu_{jj}} > 0$, for all $j \in \{0,1,\dots\}$
  - Otherwise, either all states are transient or all states are null recurrent:
    - In either case, no stationary probability vector exists
- From above two theorems, it follows that if an irreducible DTMC is positive recurrent, then fraction of time for which process is in state $j$ is:
  - $\pi_j = \frac{1}{\mu_{jj}}$, where $\pi_j$ is element $j$ of vector $\pi$, which is the unique stationary probability vector

# Example

- Recall the brand-switching example we discussed earlier:
  - $X_n$ is brand chosen on a consumer's $n$th purchase
  - DTMC is irreducible with transition probability matrix $P$
- Suppose there are a total of $c < \infty$ brands:
  - so, the DTMC is finite and hence positive recurrent
- Market share of brand $j$:
  - will be equal to $\pi_j$, element $j$ of vector $\pi$, which is the unique stationary probability vector
- First, consider symmetric case where:
  - $P_{jj} = \frac{1}{2}$, for all $j$,
  - $P_{ij} = \frac{1}{2(c-1)}$, for all $i \neq j$
- **Exercise**: Show that $\pi = \left[\frac{1}{c}, \frac{1}{c}, \dots, \frac{1}{c}\right]$ satisfies $\pi = \pi P$

# Example (contd.)

- Now, suppose manufacturer of brand $x$ wishes to increase that brand's market share by increasing $P_{xx}$
- Suppose:
  - ❑ $P_{xj} = \frac{1-P_{xx}}{c-1}$ for all $j \neq x$
- For all $i \neq x$, assume that as before:
  - ❑ $P_{ii} = \frac{1}{2}$,
  - ❑ $P_{ij} = \frac{1}{2(c-1)}$, for all $j \neq i$
- Consider a new two-state DTMC, where:
  - ❑ state $0$ denotes brand $x$, and
  - ❑ state $1$ denotes *not* brand $x$
- This is a two-state DTMC with transition probability matrix:
  - ❑ $P = \begin{bmatrix} P_{xx} & 1-P_{xx} \\ \frac{1}{2(c-1)} & \frac{(2c-3)}{2(c-1)} \end{bmatrix}$
- Solution of $\pi = \pi P$ along with $\pi_0 + \pi_1 = 1$ gives the market share of brand $x$ as:
  - ❑$\pi_0 = \frac{1}{(2c-2)(1-P_{xx})+1}$
- $\pi_0$:
  - ❑equals $\frac{1}{c}$ as before if $P_{xx} = \frac{1}{2}$
  - ❑increases in $P_{xx}$
  - ❑$\rightarrow 1$ as $P_{xx} \rightarrow 1$
- **Note**:
  - ❑Usually, combining states as we have done destroys Markovian property, i.e., we would no longer have a DTMC
  - ❑However, this technique works here because of the specific symmetrical structure of the state transition diagram

# Example

- Consider a DTMC with an infinite number of states
- Let state space be $\{0,1,2,\dots\}$ and transition probability matrix be:
  - $P = \begin{bmatrix} 0 & 1 & 0 & \dots & & \\ q & 0 & p & 0 & \dots & \\ 0 & q & 0 & p & 0 & \dots \\ & \vdots & & \vdots & & \vdots \end{bmatrix}$,
  - where $qp > 0$ and $q + p = 1$
- Is this DTMC irreducible?
  - Yes
- System of equations $\pi = \pi P$ becomes:
  - $\pi_0 = q\pi_1$,
  - $\pi_1 = \pi_0 + q\pi_2$,
  - $\pi_j = p\pi_{j-1} + q\pi_{j+1}, j \in \{2,3,\dots\}$
- Solving this system recursively, we get:
  - $\pi_1 = \frac{\pi_0}{q}$,
  - $\pi_{j+1} = \frac{p\pi_j}{q}, j \in \{1,2,\dots\}$
- $\pi_j$ in terms of $\pi_0$:
  - $\pi_j = \left(\frac{p}{q}\right)^{j-1} \frac{\pi_0}{q}, j \in \{1,2,\dots\}$
- $\sum_{j=0}^{\infty} \pi_j$:
  - *1)* $\pi_0 \left[1 + \sum_{j=1}^{\infty} \frac{1}{q} \left(\frac{p}{q}\right)^{j-1}\right]$
- The infinite sum in 1) converges iff $p < q$
- Hence, when $p < q$, all states are positive recurrent, and stationary distribution given by:
  - $\pi_0 = \frac{(q-p)}{2q}$
  - $\pi_j = \left(\frac{p}{q}\right)^{j-1} \frac{(q-p)}{2q^2}, j \in \{1,2,\dots\}$
- Can be shown that:
  - all states are transient if $p > q$
  - all states are null recurrent if $p = q$

# Positive Recurrence of Irreducible DTMCs (contd.)

- Recall:
  - **Theorem**:
    - For an irreducible DTMC, all states are positive recurrent iff there is a unique stationary probability vector $\pi$ satisfying $\pi = \pi P$:
      - where $\pi_j = \frac{1}{\mu_{jj}} > 0$, for all $j \in \{0,1, \dots\}$
    - Otherwise, either all states are transient or all states are null recurrent:
      - In either case, no stationary probability vector exists
- Suppose:
  - $P$ is the transition probability matrix of an irreducible DTMC
  - $\pi$ is a vector with non-negative elements such that:
    - $\pi = \pi P$, and
    - $\pi_i > 0$, for *some* $i \in \{0,1, \dots\}$
- Consider any $j \in \{0,1, \dots\}$
- Since the DTMC is irreducible, there exists $m \geq 1$, s.t. $P_{ij}^m > 0$
- Also, $\pi = \pi P^m$, and hence:
  - $\pi_j = \sum_{l=0}^{\infty} \pi_l P_{lj}^m > 0$
- Hence, *every* element of $\pi$ is positive
- Also, note that for every $\alpha > 0$:
  - $(\alpha\pi) = (\alpha\pi)P$
- Hence, $\pi$ can be normalized so that its elements constitute a PMF on $\{0,1, \dots\}$

# Positive Recurrence of Irreducible DTMCs (contd.)

- Recall:
  - ❑ **Theorem**:
    - For an irreducible DTMC, all states are positive recurrent iff there is a unique stationary probability vector $\pi$ satisfying $\pi = \pi P$:
      - where $\pi_j = \frac{1}{\mu_{jj}} > 0$, for all $j \in \{0,1, \dots\}$
    - Otherwise, either all states are transient or all states are null recurrent:
      - In either case, no stationary probability vector exists
- To prove the "if" part, suppose there exists a unique stationary probability vector $\pi$ satisfying $\pi = \pi P$
- To reach a contradiction, suppose the states are transient or null recurrent
- Then for every $i$ and $j$:

  *1)* $\lim_{n\to\infty} \frac{\sum_{k=1}^{n} P_{ij}^{k}}{n} = 0$
- Let $\pi$ be a probability vector s.t. $\pi = \pi P$
- Then for every $k \geq 1$:
  - ❑ $\pi = \pi P^k$

2) So, $\pi = \pi \frac{1}{n} \sum_{k=1}^{n} P^k$

- Taking limits as $n \to \infty$, expression on RHS of 2) converges to $0$, whereas $\pi_j > 0$, for all $j \in \{0,1, \dots\}$, which is a contradiction
- Hence, the states of the DTMC are positive recurrent

# Positive Recurrence of Irreducible DTMCs (contd.)

- Recall:
  - **Theorem**:
    - For an irreducible DTMC, all states are positive recurrent iff there is a unique stationary probability vector $\pi$ satisfying $\pi = \pi P$:
      - where $\pi_j = \frac{1}{\mu_{jj}} > 0$, for all $j \in \{0,1,\dots\}$
    - Otherwise, either all states are transient or all states are null recurrent:
      - In either case, no stationary probability vector exists
- To prove the "only if" part, suppose the states are positive recurrent
- As before, for $s \geq 0$, let $P(X_s = i) = a_i^{(s)}$
- Define:

  *1)* $b^{(n)} = \frac{1}{n}\sum_{s=1}^{n} a^{(s)}$, and

  *2)* $c^{(n)} = \frac{1}{n}\sum_{s=1}^{n} a^{(s-1)}$
- Then $b^{(n)}$ and $c^{(n)}$ are probability vectors

- Recall: $a^{(s)} = aP^s$, where $a_i = P(X_0 = i)$
- So, $b^{(n)} = a\frac{1}{n}\sum_{s=1}^{n} P^s$, i.e.:

  *3)* $b_j^{(n)} = \sum_{i=0}^{\infty} a_i \frac{\sum_{s=1}^{n} P_{ij}^s}{n}$
- Since all states are positive recurrent:

  *4)* $\lim_{n\to\infty} \frac{\sum_{s=1}^{n} P_{ij}^s}{n} = \frac{1}{\mu_{jj}} > 0$
- By 3) and 4), $\lim_{n\to\infty} b_j^{(n)}$:
  - $\sum_{i=0}^{\infty} a_i \lim_{n\to\infty}\left(\frac{\sum_{s=1}^{n} P_{ij}^s}{n}\right) = \frac{1}{\mu_{jj}}$
  - where the exchange of $\lim_{n\to\infty}(\cdot)$ and $\sum_{i=0}^{\infty}(\cdot)$ is valid due to the bounded convergence theorem (since $\frac{\sum_{s=1}^{n} P_{ij}^s}{n} \leq 1$)
- That is:

  *5)* $\lim_{n\to\infty} b^{(n)} = \left[\frac{1}{\mu_{00}}, \frac{1}{\mu_{11}}, \dots\right] = \pi$ (say)
- Similar to proof of 5), we get:

  6) $\lim_{n\to\infty} c^{(n)} = \pi$

# Positive Recurrence of Irreducible DTMCs (contd.)

- Recall:
  - **Theorem**:
    - For an irreducible DTMC, all states are positive recurrent iff there is a unique stationary probability vector $\pi$ satisfying $\pi = \pi P$:
      - where $\pi_j = \frac{1}{\mu_{jj}} > 0$, for all $j \in \{0,1,\dots\}$
    - Otherwise, either all states are transient or all states are null recurrent:
      - In either case, no stationary probability vector exists

*1)* $b^{(n)} = \frac{1}{n}\sum_{s=1}^{n} a^{(s)}$, and

*2)* $c^{(n)} = \frac{1}{n}\sum_{s=1}^{n} a^{(s-1)}$

*5)* $\lim_{n\to\infty} b^{(n)} = \left[\frac{1}{\mu_{00}}, \frac{1}{\mu_{11}}, \dots\right] = \pi$

*6)* $\lim_{n\to\infty} c^{(n)} = \pi$

- By 1) and 2), we get:

*7)* $b^{(n)} = c^{(n)} P$

- Taking limits on both sides of 7), and using 5) and 6), we get:

*8)* $\pi = \pi P$

- Thus, there exists a solution to system of equations $\pi = \pi P$, all of whose elements are positive
- Next, we will prove that there exists a solution to the system:

*9)* $\tilde{\pi} = \tilde{\pi} P$, $\tilde{\pi}_i > 0$, $\forall i$, and $\sum_{i=0}^{\infty} \tilde{\pi}_i = 1$

- Recall: $b^{(n)}$ is a probability vector $\forall n$
- Hence, $\forall n$:

*10)* $\sum_{i=0}^{\infty} b_i^{(n)} = 1$

- However, 5) and 10) do not immediately imply $\sum_{i=0}^{\infty} \pi_i = 1$ since that would require following exchange of limit with infinite summation to be valid:

*11)* $\sum_{i=0}^{\infty} \lim_{n\to\infty} b_i^{(n)} = \lim_{n\to\infty} \sum_{i=0}^{\infty} b_i^{(n)}$

- However, such an exchange is not always valid:
  - in general, limit of sequence of probability distributions may not be probability distribution
  - e.g., $\tilde{b}_i^{(n)} = \begin{cases} 1, & \text{if } i = n, \\ 0, & \text{else.} \end{cases}$
- So, to prove 9), note that for each $k \geq 1$:
  - $0 < \sum_{i=0}^{k} \pi_i = \sum_{i=0}^{k} \lim_{n\to\infty} b_i^{(n)} = \lim_{n\to\infty} \sum_{i=0}^{k} b_i^{(n)} \leq 1$
- Hence, $0 < \sum_{i=0}^{\infty} \pi_i \leq 1$
- Let $\alpha = \sum_{i=0}^{\infty} \pi_i > 0$
- Then $\frac{\pi}{\alpha} = \left(\frac{\pi}{\alpha}\right) P$ and $\frac{\pi}{\alpha}$ is a probability vector with all elements positive, which proves that there exists a solution to system in 9)

# Positive Recurrence of Irreducible DTMCs (contd.)

- Recall:
  - **Theorem**:
    - For an irreducible DTMC, all states are positive recurrent iff there is a unique stationary probability vector $\pi$ satisfying $\pi = \pi P$:
      - where $\pi_j = \frac{1}{\mu_{jj}} > 0$, for all $j \in \{0,1,\dots\}$
    - Otherwise, either all states are transient or all states are null recurrent:
      - In either case, no stationary probability vector exists
  - There exists a solution to the system:

    *9)* $\tilde{\pi} = \tilde{\pi} P$, $\tilde{\pi}_i > 0$, $\forall i$, and $\sum_{i=0}^{\infty} \tilde{\pi}_i = 1$
- We now prove that the system in 9) has a unique solution
- Let $\pi$ be any solution of the system in 9)
- Then for every $k \geq 1$:
  - $\pi = \pi P^k$
- Hence:
  - $\pi = \pi \frac{1}{n} \sum_{k=1}^{n} P^k$
- Taking limits as $n \to \infty$, exactly as limit of $b^{(n)}$ was taken above, and using the fact that $\pi$ is a probability vector, we get:

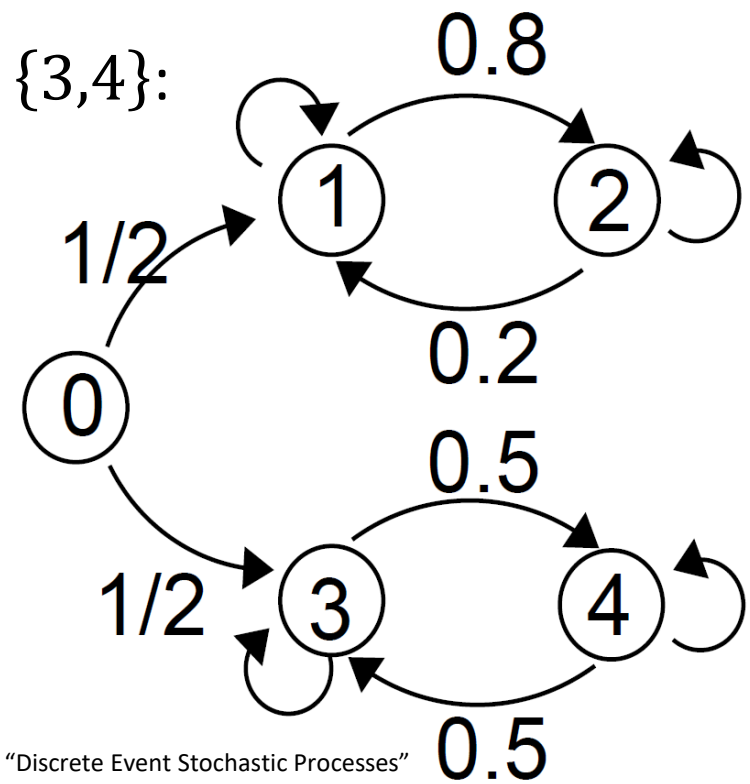
  *10)* $\pi = \left[\frac{1}{\mu_{00}}, \frac{1}{\mu_{11}}, \dots\right]$
- Hence, the system in 9) has a unique solution
- Now, suppose all states are transient or null recurrent
- If a stationary probability vector $\pi$ exists, then as before, 10) holds
- But $\mu_{jj} = \infty$, $\forall j$; so, $\pi = 0$, which contradicts the assumption that it is a probability vector
- Hence, no stationary probability vector exists

# Multiple Positive Recurrent Classes

- Consider a DTMC with multiple positive recurrent communication classes $\mathcal{C}_1, \ldots, \mathcal{C}_m$
- For $k \in \{1, \ldots, m\}$, restriction of DTMC to class $\mathcal{C}_k$ has a unique stationary distribution, say $a_k$
- By adding zeros where required, expand $a_k$ into probability vector over entire state space of original DTMC; call it $\hat{\pi}_k$
- Then $\hat{\pi}_k$ is a probability vector that satisfies $\hat{\pi}_k = \hat{\pi}_k P$
- Hence, any $\pi$ that is a convex combination of $\hat{\pi}_1, \ldots, \hat{\pi}_m$ is also a probability vector that satisfies $\pi = \pi P$
- Thus, the DTMC has a convex set of stationary probability vectors, whose extreme points are the vectors $\hat{\pi}_1, \ldots, \hat{\pi}_m$
- **Exercise**: Show that if $\pi$ is a stationary probability vector for an arbitrary DTMC, then the stationary probability $\pi_j = 0$ for every state $j$ that is either transient or null recurrent
- Hence, a DTMC has a unique stationary probability vector iff there is exactly one positive recurrent class

# Example

- Consider the DTMC with state transition diagram shown in Fig.
- Classes:
  - ❑ positive recurrent classes: $\{1,2\}$ and $\{3,4\}$
  - ❑ transient class: $\{0\}$
- Want to find stationary probability vectors
- Unique stationary probability vector for $\{1,2\}$:
  - ❑ $\begin{bmatrix} \frac{1}{5} & \frac{4}{5} \end{bmatrix}$
- Unique stationary probability vector for $\{3,4\}$:
  - ❑ $\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix}$
- Stationary probability vector for overall DTMC:
  - ❑ of the form $\begin{bmatrix} 0 & \frac{\alpha_1}{5} & \frac{4\alpha_1}{5} & \frac{\alpha_2}{2} & \frac{\alpha_2}{2} \end{bmatrix}$,
  - ❑ where $\alpha_1, \alpha_2 \geq 0$ and $\alpha_1 + \alpha_2 = 1$




Ref: A. Kumar, "Discrete Event Stochastic Processes"

# Transition Rates Between States and Classes

- Recall: for an irreducible and positive recurrent DTMC:
  - $\pi_j$ is average number of visits to state $j$ per unit time
- Clearly, if DTMC visits (arrives at) state $j$ at time $n$, then it must leave (depart from) state $j$ at time $n+1$
  - Note that a transition $j \to j$ counts as both a visit and a departure
- Let $A_j(n)$ (respectively, $D_j(n)$) be number of visits to (respectively, departures from) state $j$ in the interval $\{1, \dots, n\}$
- Since visits and departures alternate:

  *1)* $\left|D_j(n) - A_j(n)\right| \leq 1, \forall n \in \{1,2, \dots\}$
- Dividing both sides of 1) by $n$ and taking limits as $n \to \infty$, we get:

  2) $\text{transition rate out of state } j = \text{transition rate into state } j$
- Average number of transitions of type $i \to j$ per unit time:

  *3)* $\pi_i P_{ij}$

# Transition Rates Between States and Classes (contd.)

- Recall:

  2) $\text{transition rate out of state } j = \text{transition rate into state } j$

  - Average number of transitions of type $i \to j$ per unit time:

    *3)* $\pi_i P_{ij}$

- Summing 3) over all states $i$, transition rate into state $j$:
  - $\sum_i \pi_i P_{ij}$
- Thus:

  *4)* $\pi_j = \sum_i \pi_i P_{ij}$ ,

  - which we also know from the fact that $\pi = \pi P$
- Similarly, if we partition state space of DTMC into two disjoint subsets $A$ and $A^c$:
  - $\text{transition rate from } A \text{ to } A^c = \text{transition rate from } A^c \text{ to } A$
  - i.e., $\sum_{j \in A} \sum_{i \in A^c} \pi_j \, P_{ji} = \sum_{j \in A^c} \sum_{i \in A} \pi_j \, P_{ji}$

# Example

- Suppose a production process changes states in accordance with an irreducible and positive recurrent DTMC with transition probabilities $P_{ij}, i, j \in \{1, \dots, n\}$
- Let:
  - $A$: set of acceptable states,
  - $A^c$: set of unacceptable states
- Process said to be up (respectively, down) when in an acceptable (respectively, unacceptable) state
- Assume that the process with lumped states $\{A, A^c\}$ is also a DTMC
- Rate at which process goes from up to down (i.e., rate at which breakdowns occur):

  *1)* $\sum_{i \in A} \sum_{j \in A^c} \pi_i \, P_{ij}$
- Want to find:
  - $\overline{U}$: average amount of time process remains up when it goes up
  - $\overline{D}$: average amount of time process remains down when it goes down
- However, since there is a single breakdown every (up time + down time) cycle, rate at which breakdowns occur:

  *2)* $\frac{1}{\overline{U}+\overline{D}}$
- By 1) and 2):

  *3)* $\frac{1}{\overline{U}+\overline{D}} = \sum_{i \in A} \sum_{j \in A^c} \pi_i \, P_{ij}$

# Example (contd.)

- Recall:

  *3)* $\frac{1}{\overline{U}+\overline{D}} = \sum_{i \in A} \sum_{j \in A^c} \pi_i \, P_{ij}$

- Fraction of time for which process is up:

  *4)* $\frac{\overline{U}}{\overline{U}+\overline{D}}$

- However, this fraction of time also equals:

  *5)* $\sum_{i \in A} \pi_i$

- By 4) and 5):

  *6)* $\frac{\overline{U}}{\overline{U}+\overline{D}} = \sum_{i \in A} \pi_i$

- Solving 3) and 6), we get:

  - ❑ $\overline{U} = \frac{\sum_{i \in A} \pi_i}{\sum_{i \in A} \sum_{j \in A^c} \pi_i P_{ij}}$
  - ❑ $\overline{D} = \frac{\sum_{i \in A^c} \pi_i}{\sum_{i \in A} \sum_{j \in A^c} \pi_i P_{ij}}$

# Pointwise Limits

- Recall: Blackwell's Theorem:
  i. If $F$ is not lattice, then for all $a \geq 0$:
     - $m(t+a) - m(t) \to \frac{a}{\mu}$ as $t \to \infty$
  ii. If $F$ is lattice with period $d$, then:
     - $E[\text{number of renewals at } nd] \to \frac{d}{\mu}$ as $n \to \infty$
- Recall: a DTMC can be viewed as a renewal process in which visits to given state $j$ are renewals
- Since DTMC is a discrete-time process, visits to state $j$ only occur at integer values of time:
  - ❑ so, corresponding renewal process has lattice interarrival distribution
- Hence, by part (ii) of Blackwell's theorem, we get following result
- **Theorem**: If states $i$ and $j$ communicate, then:
  a) If $j$ is aperiodic, then $\lim_{n\to\infty} P_{ij}^{n} = \frac{1}{\mu_{jj}}$
  b) If $j$ has period $d$, then $\lim_{n\to\infty} P_{jj}^{nd} = \frac{d}{\mu_{jj}}$

# Pointwise Limits (contd.)

- Recall:
  - **Theorem**:
    - For an irreducible DTMC, all states are positive recurrent iff there is a unique stationary probability vector $\pi$ satisfying $\pi = \pi P$:
      - where $\pi_j = \frac{1}{\mu_{jj}} > 0$, for all $j \in \{0,1,\dots\}$
    - Otherwise, either all states are transient or all states are null recurrent:
      - In either case, no stationary probability vector exists
  - **Theorem**: If states $i$ and $j$ communicate, then:

    a) If $j$ is aperiodic, then $\lim_{n\to\infty} P_{ij}^n = \frac{1}{\mu_{jj}}$

    b) If $j$ has period $d$, then $\lim_{n\to\infty} P_{jj}^{nd} = \frac{d}{\mu_{jj}}$
- Above two theorems imply the following theorem
- **Theorem**: For an irreducible and aperiodic DTMC:

  a) Either the states are all transient or all null recurrent; in this case, $\lim_{n\to\infty} P_{ij}^n = 0$ for all $i, j$, and there exists no stationary probability vector

  b) Or else, all states are positive recurrent. In this case, there is a unique stationary probability vector $\pi$ satisfying $\pi = \pi P$. Also, for all states $i$ and $j$:
     - $\lim_{n\to\infty} P_{ij}^n = \pi_j = \frac{1}{\mu_{jj}} > 0$

# Ergodic DTMCs

- Recall:
  - **Theorem**: For an irreducible and aperiodic DTMC:
    a) Either the states are all transient or all null recurrent; in this case, $\lim_{n\to\infty} P_{ij}^n = 0$ for all $i, j$, and there exists no stationary probability vector
    b) Or else, all states are positive recurrent. In this case, there is a unique stationary probability vector $\pi$ satisfying $\pi = \pi P$. Also, for all states $i$ and $j$:
       - $\lim_{n\to\infty} P_{ij}^n = \pi_j = \frac{1}{\mu_{jj}} > 0$
- **Definition**: A DTMC is said to be *ergodic* if it is irreducible, aperiodic, and positive recurrent
- Note that for an ergodic DTMC, the situation is as in part b) of above theorem

# Example: The M/G/1 Queue

- Recall earlier example: an M/G/1 queue with arrival rate $\lambda$ and service distribution $G(\cdot)$
- Let:
  - $X_n$: no. of customers left behind by $n$th departure, for $n \geq 1$
  - $a_j = \int_0^\infty e^{-\lambda x} \frac{(\lambda x)^j}{j!} dG(x)$: probability of $j$ arrivals during a service period, where $j \in \{0,1,2,\dots\}$
- We showed that $\{X_n, n = 1,2,\dots\}$ is a DTMC with following transition probabilities:
  - $P_{0j} = a_j$, for $j \in \{0,1,2,\dots\}$
  - $P_{ij} = \begin{cases} a_{j-i+1}, & \text{if } j \geq i-1, i \geq 1, \\ 0, & \text{else.} \end{cases}$
- Mean number of arrivals during a service period:
  - $\rho = \sum_j j\, a_j$
- Conditioning on length of service period:
  - $\rho = \lambda E[S]$, where $S$ is a service period and has distribution $G(\cdot)$

# Example: The M/G/1 Queue (contd.)

- Recall:
  - $P_{0j} = a_j$, for $j \in \{0,1,2,\dots\}$
  - $P_{ij} = \begin{cases} a_{j-i+1}, & \text{if } j \geq i-1, i \geq 1, \\ 0, & \text{else.} \end{cases}$
  - $\rho = \sum_j j\, a_j = \lambda E[S]$
- The system of equations $\pi = \pi P$ becomes:

  *1)* $\pi_j = \pi_0 a_j + \sum_{i=1}^{j+1} \pi_i a_{j-i+1}$, for $j \in \{0,1,2,\dots\}$
- Introduce the following generating functions:
  - $\pi(s) = \sum_{j=0}^{\infty} \pi_j s^j$
  - $A(s) = \sum_{j=0}^{\infty} a_j s^j$
- From 1), we get $\pi(s)$:
  - $\pi_0 A(s) + \sum_{j=0}^{\infty} \sum_{i=1}^{j+1} \pi_i a_{j-i+1}\, s^j$
  - $\pi_0 A(s) + s^{-1} \sum_{i=1}^{\infty} \pi_i s^i \sum_{j=i-1}^{\infty} a_{j-i+1}\, s^{j-i+1}$
  - $\pi_0 A(s) + (\pi(s) - \pi_0) \frac{A(s)}{s}$

# Example: The M/G/1 Queue (contd.)

- Recall:
  - $\rho = \sum_j j\, a_j = \lambda E[S]$
  - $\pi(s) = \pi_0 A(s) + (\pi(s) - \pi_0)\frac{A(s)}{s}$
- Solving for $\pi(s)$, we get:

  *2)* $\pi(s) = \frac{(s-1)\pi_0 A(s)}{s - A(s)}$
- Taking limits as $s \to 1$, we get:
  - $\lim_{s\to 1} \pi(s) = \frac{\pi_0}{1 - A'(1)}$
  - $= \frac{\pi_0}{1-\rho}$
  - where we used L'Hôpital's rule and the facts that $\lim_{s\to 1} A(s) = 1$ and $A'(1) = \sum_{j=0}^{\infty} j a_j = \rho$
- So, $\sum_{i=0}^{\infty} \pi_i = \frac{\pi_0}{1-\rho}$
- Hence, stationary probabilities exist only when $\rho < 1$, and in this case:
  - $\pi_0 = 1 - \rho = 1 - \lambda E[S]$
  - $\pi(s) = \frac{(1-\lambda E[S])(s-1)A(s)}{s - A(s)}$

# Example: The G/M/1 Queue

- Recall earlier example: G/M/1 queue with interarrival distribution $G(\cdot)$ and service rate $\mu$
- Let $X_n$: number of customers in system as seen by $n$th arrival, for $n \geq 1$
- We showed that $\{X_n, n = 1,2, \dots\}$ is a DTMC with following transition probabilities:
  - ❑ $P_{i,i+1-j} = \int_0^\infty e^{-\mu t} \frac{(\mu t)^j}{j!} dG(t)$, for $j \in \{0,1, \dots, i\}$
  - ❑ $P_{i,0} = \int_0^\infty \left( \sum_{k=i+1}^\infty e^{-\mu t} \frac{(\mu t)^k}{k!} \right) dG(t)$ , for $i \in \{0,1, \dots\}$
- The system of equations $\pi = \pi P$ becomes:

  *1)* $\pi_k = \sum_{i=k-1}^\infty \pi_i \int_0^\infty e^{-\mu t} \frac{(\mu t)^{i+1-k}}{(i+1-k)!} dG(t)$, for $k \in \{1,2, \dots\}$
  - ❑ (we have excluded the equation $\pi_0 = \sum_i \pi_i P_{i0}$ since one equation in the system $\pi = \pi P$ is always redundant)

# Example: The G/M/1 Queue (contd.)

- Recall: the system of equations $\pi = \pi P$ becomes:

  *1)* $\pi_k = \sum_{i=k-1}^{\infty} \pi_i \int_0^{\infty} e^{-\mu t} \frac{(\mu t)^{i+1-k}}{(i+1-k)!} dG(t)$, for $k \in \{1,2,\dots\}$
- We try solution of the form $\pi_k = c\beta^k$, for $k \in \{0,1,\dots\}$
- Substituting into 1) and simplifying, we get:
  - $c\beta^k = c \int_0^{\infty} e^{-\mu t} \beta^{k-1} \left( \sum_{i=k-1}^{\infty} \frac{(\beta\mu t)^{i+1-k}}{(i+1-k)!} \right) dG(t)$
  - $= c \int_0^{\infty} e^{-\mu t} \beta^{k-1} e^{\beta\mu t} dG(t)$
- So:

  *2)* $\beta = \int_0^{\infty} e^{-\mu t(1-\beta)} dG(t)$
- Using $\sum_{i=0}^{\infty} \pi_i = 1$, we get:
  - $c = 1 - \beta$
- Can be shown that if the mean of $G(\cdot)$ is greater than the mean service time $\frac{1}{\mu}$, then there is a unique value of $\beta$ satisfying 2) that is between 0 and 1 (this value can be found by numerical methods)
- In this case, since $\pi_k$ are the *unique* solutions to 1) and $\sum_{i=0}^{\infty} \pi_i = 1$, and above guessed solution satisfies them, it follows that:
  - $\pi_k = (1-\beta)\beta^k$, for $k \in \{0,1,\dots\}$

# Time-Reversible DTMCs

- Consider a DTMC with transition probabilities $P_{ij}$ and stationary probabilities $\pi_i$
- Assume that $P\{X_n = i\} = \pi_i$, for all $i, n$
- Suppose that starting at some time $n$, we trace the sequence of states going backwards in time, $X_n, X_{n-1}, \ldots$
- This sequence of states is itself a DTMC with following transition probabilities:
  - $P_{ij}^* = P\{X_m = j | X_{m+1} = i\}$
  - $= \frac{P\{X_{m+1} = i | X_m = j\} P\{X_m = j\}}{P\{X_{m+1} = i\}}$
  - $= \frac{\pi_j P_{ji}}{\pi_i}$
- To prove that reversed process is a DTMC, note that:

  *1)* $P\{X_m = j | X_{m+1} = i, X_{m+2}, X_{m+3}, \ldots\} = P\{X_m = j | X_{m+1} = i\}$
  - reason: since $X_n, n \geq 1$ is a DTMC, it follows that given the present state $X_{m+1}$, the past state $X_m$ and future states $X_{m+2}, X_{m+3}, \ldots$ are independent
- If $P_{ij}^* = P_{ij}$ for all $i, j$, then the DTMC is said to be time reversible
- That is, DTMC is time reversible if:

  *2)* $\pi_i P_{ij} = \pi_j P_{ji}$, for all $i, j$
- *Interpretation*:
  - Equation 2) says that transition rate from $i$ to $j$ equals transition rate from $j$ to $i$
  - necessary for time reversibility since a transition from $i$ to $j$ going backward in time is equivalent to a transition from $j$ to $i$ going forward in time

# Time-Reversible DTMCs (contd.)

- Recall: DTMC is time reversible if:

  *2)* $\pi_i P_{ij} = \pi_j P_{ji}$, for all $i, j$

- Suppose the DTMC is irreducible
- If we can find non-negative numbers $x_i$, summing to 1, which satisfy 2), then:
  - ❑the DTMC is time reversible and the numbers represent the stationary probabilities
- **Proof**:
  - ❑Suppose $x_i P_{ij} = x_j P_{ji}$, for all $i, j$ and $\sum_i x_i = 1$
  - ❑Summing the equation $x_i P_{ij} = x_j P_{ji}$ over all $i$ gives:

    *3)* $\sum_i x_i P_{ij} = x_j$

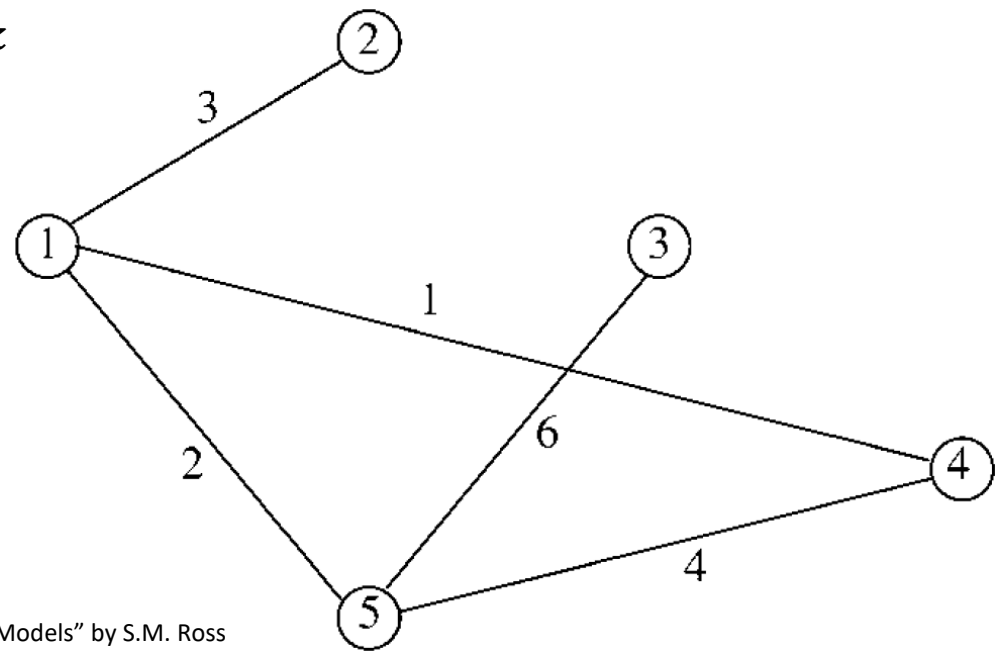
  - ❑So, the numbers $x_i$ are the unique solution to 3) and $\sum_i x_i = 1$, i.e., stationary probabilities

# Example 1

- Consider a random walk with states $0,1,\ldots,M$ and transition probabilities:
  - $P_{i,i+1} = \alpha_i = 1 - P_{i,i-1}$, for $i \in \{1,\ldots M-1\}$
  - $P_{0,1} = \alpha_0 = 1 - P_{0,0}$,
  - $P_{M,M} = \alpha_M = 1 - P_{M,M-1}$
- Without any need for computations, we can argue that this DTMC is time reversible
- This is because in all time intervals:
  - $\forall i \in \{0,\ldots M-1\}$, number of transitions from $i$ to $i+1$ is within one of number of transitions from $i+1$ to $i$
- So:
  - rate of transitions from $i$ to $i+1$ equals rate of transitions from $i+1$ to $i$
- Hence, the DTMC is time reversible

# Example 2

- Consider an arbitrary undirected connected graph (e.g.. shown in fig.):
  - a non-negative weight $w_{ij}$ is associated with each edge $(i, j)$
- A particle performs random walk on this graph as follows:
  - if at any time, the particle is at node $i$, then it will next move to node $j$ w.p. $P_{ij} = \frac{w_{ij}}{\sum_k w_{ik}}$
  - e.g., in the graph in fig., $P_{12} = \frac{3}{3+1+2} = \frac{1}{2}$

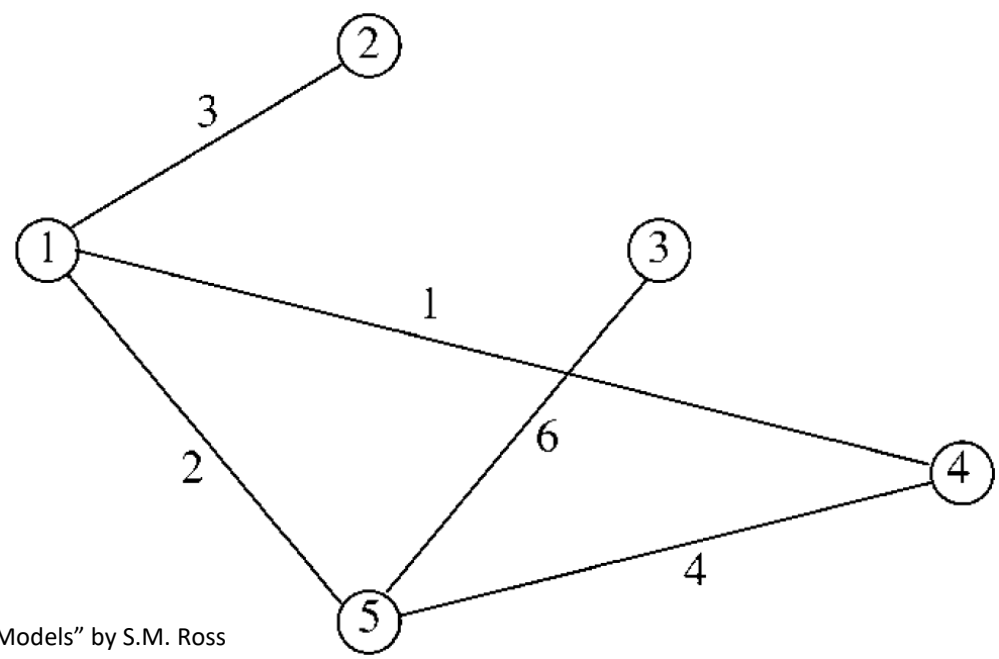



Ref: "Introduction to Probability Models" by S.M. Ross

# Example 2 (contd.)

- Recall:
  - $P_{ij} = \frac{w_{ij}}{\sum_k w_{ik}}$
- Time reversibility condition $\pi_i P_{ij} = \pi_j P_{ji}$ becomes:
  - $\frac{\pi_i}{\sum_k w_{ik}} = \frac{\pi_j}{\sum_k w_{jk}}$
- Solution:
  - $\pi_i = c \sum_k w_{ik}$
- Using $\sum_i \pi_i = 1$, we get:

  *1)* $\pi_i = \frac{\sum_k w_{ik}}{\sum_k \sum_l w_{kl}}$
- Hence, process is reversible with stationary probabilities given by 1)




Ref: "Introduction to Probability Models" by S.M. Ross

# Example 3

- Recall earlier example:
  - ❑ There is a DTMC with an infinite number of states
  - ❑ State space is $\{0,1,2,\dots\}$ and transition probability matrix is:
    - $P = \begin{bmatrix} 0 & 1 & 0 & \dots & & \\ q & 0 & p & 0 & \dots & \\ 0 & q & 0 & p & 0 & \dots \\ & \vdots & & \vdots & & \vdots \end{bmatrix}$,
    - where $qp > 0$ and $q + p = 1$
- We showed that when $p < q$, all states are positive recurrent and stationary distribution is given by:

  *1)* $\pi_0 = \frac{(q-p)}{2q}$

  *2)* $\pi_j = \left(\frac{p}{q}\right)^{j-1} \frac{(q-p)}{2q^2}, j \in \{1,2,\dots\}$
- Time reversibility condition $\pi_i P_{ij} = \pi_j P_{ji}$ becomes:

  *3)* $\pi_0 = q\pi_1$,

  *4)* $p\pi_i = q\pi_{i+1}$, for $i \in \{1,2,\dots\}$
- The probabilities $\pi_i$ in 1) and 2) satisfy 3) and 4)
- Hence, the process is time reversible

# Kolmogorov's Criterion

- To prove the condition $\pi_i P_{ij} = \pi_j P_{ji}$, for all $i, j$, we need to know the stationary probabilities
- Now, we discuss a result, which can be used to prove reversibility without knowing the stationary probabilities
- **Theorem**: A stationary DTMC is time reversible iff, the probability of traversing any loop is equal to the probability of traversing the same loop in the reverse direction, i.e., if:

  *1)* $P_{i,i_1} P_{i_1,i_2} \dots P_{i_k,i} = P_{i,i_k} P_{i_k,i_{k-1}} \dots P_{i_1,i}$, for all states $i, i_1, \dots, i_k$
- **Proof**:
  - ❑To prove necessity, note that:
    - ○$\pi_i P_{i,i_1} = \pi_{i_1} P_{i_1,i}$
    - ○$\pi_{i_1} P_{i_1,i_2} = \pi_{i_2} P_{i_2,i_1}$
    - ○$\vdots$
    - ○$\pi_{i_k} P_{i_k,i} = \pi_i P_{i,i_k}$
  - ❑Multiplying preceding equations and cancelling stationary probabilities, we get 1)
  - ❑To prove sufficiency, fix states $i$ and $j$
  - ❑From 1), we get:

    *2)* $P_{i,i_1} P_{i_1,i_2} \dots P_{i_k,j} P_{j,i} = P_{i,j} P_{j,i_k} \dots P_{i_1,i}$, for all states $i_1, \dots, i_k$
  - ❑ Summing 2) over all states $i_1, \dots, i_k$, we get:
    - ○$P_{ij}^{k+1} P_{ji} = P_{ij} P_{ji}^{k+1}$
  - ❑ Hence:
    - ○$P_{ji} \frac{\sum_{k=1}^{n} P_{ij}^{k+1}}{n} = P_{ij} \frac{\sum_{k=1}^{n} P_{ji}^{k+1}}{n}$
  - ❑ Taking limits as $n \to \infty$, we get $P_{ji} \pi_j = P_{ij} \pi_i$

# Example

- Suppose there is a list of elements numbered $1, \ldots, n$
- Consider a DTMC with $n!$ states, with the state at any time being:
  - the order in which the $n$ elements are listed
- Suppose $P_1, \ldots, P_n$ are non-negative numbers that add up to $1$
- At each time:
  - one of the $n$ elements is selected at random, with element $i$ being selected w.p. $P_i$, and
  - selected element is swapped with element just before it in the list:
    - except when the first element in the list is selected, in which case no change is made
- E.g.:
  - Suppose present list ordering is $1,3,4,2,5$
  - Element $2$ is selected
  - Then new ordering becomes:
    - $1,3,2,4,5$
- **Proposition**: Above DTMC is time reversible

# Example (contd.)

- Recall:
  - **Proposition**: Above DTMC is time reversible
- This is because starting in any state $x$, probability of traversing any loop is equal to probability of traversing same loop in reverse direction
- E.g.,
  - Suppose $n = 3$ and consider the following loop from state $(1,2,3)$ to itself:
    - $(1,2,3) \rightarrow (2,1,3) \rightarrow (2,3,1) \rightarrow (3,2,1) \rightarrow (3,1,2) \rightarrow (1,3,2) \rightarrow (1,2,3)$
  - Product of transition probabilities in forward direction:
    - $P_2 P_3 P_3 P_1 P_1 P_2 = P_1^2 P_2^2 P_3^2$
  - Product of transition probabilities in reverse direction:
    - $P_3 P_3 P_2 P_2 P_1 P_1 = P_1^2 P_2^2 P_3^2$
- To prove above claim in general, let $f_i$ be number of times element $i$ moves forward while traversing loop in first direction
- Since the loop goes from a fixed state back to itself, it follows that element $i$ also moves backward $f_i$ times
- Hence, product of transition probabilities for both forward and reverse loop traversal will equal:
  - $\prod_i P_i^{f_i}$

# Using Reversed Chain When Process is Not Time Reversible

- **Theorem**:
  - Consider an irreducible DTMC with transition probabilities $P_{ij}$
  - Suppose one can find non-negative numbers $\pi_i$, $i \geq 0$, summing to unity, and a transition probability matrix with elements $Q_{ij}$ such that for all $i, j$:
    1) $\pi_i P_{ij} = \pi_j Q_{ji}$
  - Then $\pi_i$, $i \geq 0$, are the stationary probabilities and $Q_{ij}$ are the transition probabilities of the reversed DTMC
- **Proof**:
  - Summing 1) over all $i$ yields:
    - $\sum_i \pi_i P_{ij} = \pi_j$
  - Hence, $\pi_i$, $i \geq 0$, are the stationary probabilities of the forward DTMC
  - Since $Q_{ji} = \frac{\pi_i P_{ij}}{\pi_j}$, it follows that $Q_{ij}$ are the transition probabilities of the reversed DTMC
- Importance of above theorem is that we can sometimes guess at nature of reversed DTMC and then use the set of equations 1) to obtain both the stationary probabilities and the $Q_{ij}$

# Example

- A single bulb is necessary to light a given room
- When the bulb in use fails, it is replaced by a new one at the beginning of next day
- Each bulb, independently, fails on its $i$th day of use w.p. $p_i$, $i \geq 1$:
  - i.e., $P\{L = i\} = p_i$, where $L$ is lifetime of bulb
- Let $X_n = i$ if bulb in use at beginning of day $n$ is in its $i$th day of use
- Then $\{X_n, n \geq 1\}$ is a DTMC whose transition probabilities are as follows:
  - $P_{i,1} = P\{\text{bulb, on its } i\text{th day of use, fails}\}$
  - $= P\{\text{lifetime of bulb} = i | \text{lifetime of bulb} \geq i\}$
  - $= \frac{P\{L=i\}}{P\{L \geq i\}}$
- Also, $P_{i,i+1} = 1 - P_{i,1}$

# Example (contd.)

- Recall:
  - $P_{i,1} = \frac{P\{L=i\}}{P\{L \geq i\}}$,
  - $P_{i,i+1} = 1 - P_{i,1}$
- Suppose this DTMC is in steady state and consider the sequence of states going backward in time
- Reverse DTMC will:
  - decrease by 1 until it reaches 1
  - and then it will jump to a random value representing the lifetime of the (in real time) previous bulb
- Thus, it seems that reverse DTMC should have transition probabilities given by:
  - $Q_{i,i-1} = 1$, for $i > 1$
  - $Q_{1,i} = p_i$, for $i \geq 1$

# Example (contd.)

- Recall: it seems that reverse DTMC should have transition probabilities given by:
  - ❑ $Q_{i,i-1} = 1$, for $i > 1$
  - ❑ $Q_{1,i} = p_i$, for $i \geq 1$
- To verify this, and at the same time, determine stationary probabilities, we need to see if we can find positive numbers $\pi_i$ such that for all $i, j$:

  *1)* $\pi_i P_{ij} = \pi_j Q_{ji}$
- For $j = 1$, 1) becomes:
  - ❑ $\pi_i \frac{P\{L=i\}}{P\{L\geq i\}} = \pi_1 P(L = i)$

  2) i.e., $\pi_i = \pi_1 P\{L \geq i\}$
- Summing over all $i$, equating it to 1, and simplifying, gives:

  *3)* $\pi_1 = \frac{1}{E[L]}$
- By 2) and 3), we get:
  - ❑ $\pi_i = \frac{P\{L\geq i\}}{E[L]}$, for $i \geq 1$
- Substituting above values of $\pi_i$ and $Q_{ji}$ into the equation $\pi_i P_{i,i+1} = \pi_{i+1} Q_{i+1,i}$, we get:
  - ❑ $\frac{P\{L\geq i\}}{E[L]}\left(1 - \frac{P\{L=i\}}{P\{L\geq i\}}\right) = \frac{P\{L\geq i+1\}}{E[L]}$,
  - ❑ which is true

# Semi-Markov Processes

- Semi-Markov process is one that changes states in accordance with a DTMC, but takes a random amount of time between changes
- **Definition**: A semi-Markov process is a stochastic process, $\{Z(t), t \geq 0\}$, with states $0,1, \ldots$, which is s.t. whenever it enters state $i$, $i \geq 0$:
  a) Next state it will enter is state $j \geq 0$ w.p. $P_{ij}$
  b) Given that next state to be entered is state $j$, the time until the transition from $i$ to $j$ occurs has CDF $F_{ij}$
- Does a semi-Markov process have the Markovian property that given the present state, the future evolution is independent of the past?
  - ❑ No, since to predict the future evolution, we also need to know amount of time that has been spent in current state
- A DTMC is a semi-Markov process in which:
  - ❑ $F_{ij}(t) = \begin{cases} 0, & \text{if } t < 1, \\ 1, & \text{if } t \geq 1. \end{cases}$
  - ❑ i.e., all transition times of a DTMC are identically 1

# Semi-Markov Processes (contd.)

- Let $H_i$ be CDF of time that the semi-Markov process spends in state $i$ before making a transition
- Conditioning on next state, we get:
  - $H_i(t) = \sum_j P_{ij} F_{ij}(t)$
- Let $\mu_i$ be mean of this CDF
- Let $X_n$ denote the $n$th state visited; then $\{X_n, n \geq 0\}$ is a DTMC with transition probabilities $P_{ij}$:
  - called embedded DTMC of the semi-Markov process
- Semi-Markov process is said to be irreducible if the embedded DTMC is irreducible

# Semi-Markov Processes (contd.)

- Let:
  - $T_{ii}$ denote the time between successive transitions into state $i$
  - $\mu_{ii} = E[T_{ii}]$
- **Proposition**: If the semi-Markov process is irreducible and if $T_{ii}$ has a non-lattice CDF with finite mean, then:
  - $P_i = \lim_{t \to \infty} P\{Z(t) = i | Z(0) = j\}$

  exists, is independent of initial state $j$, and:
  - $P_i = \frac{\mu_i}{\mu_{ii}}$
- **Proposition**: If the semi-Markov process is irreducible and if $\mu_{ii} < \infty$, then w.p. 1:
  - $\lim_{t \to \infty} \frac{\text{amount of time in state } i \text{ during } [0,t]}{t} = \frac{\mu_i}{\mu_{ii}}$
- **Proof**:
  - Both propositions follow from fact that we have an alternating renewal process (delayed if $j \neq i$) in which cycle begins whenever process enters state $i$, process is "on" when in state $i$ and "off" when not in $i$

# Semi-Markov Processes (contd.)

- **Theorem**: Suppose the semi-Markov process is irreducible and $T_{ii}$ has a non-lattice CDF with finite mean. Also, suppose the embedded DTMC $\{X_n, n \geq 0\}$ is positive recurrent. Then:
  - $P_i = \frac{\pi_i \mu_i}{\sum_k \pi_k \mu_k}$
- **Proof**:
  - Let:
    - $Y_i(k)$: amount of time spent in state $i$ during $k$th visit to that state
    - $N_i(m)$: number of visits to state $i$ in first $m$ transitions of the semi-Markov process
  - Proportion of time in state $i$ during first $m$ transitions:
    - $P_{i,m} = \frac{\sum_{l=1}^{N_i(m)} Y_i(l)}{\sum_k \sum_{l=1}^{N_k(m)} Y_k(l)}$
    - $= \frac{\frac{N_i(m)}{m}\left(\frac{\sum_{l=1}^{N_i(m)} Y_i(l)}{N_i(m)}\right)}{\sum_k \frac{N_k(m)}{m}\left(\frac{\sum_{l=1}^{N_k(m)} Y_k(l)}{N_k(m)}\right)}$
  - Result follows by letting $m \to \infty$ and using strong law of large numbers and strong law for renewal processes

# Example

- Consider a machine that can be in one of three states:
  - Good condition,
  - Fair condition,
  - Broken down
- A machine in good condition:
  - will remain that way for a mean time $\mu_1$ and
  - will then go to either the fair condition or broken condition w.p. $\frac{3}{4}$ and $\frac{1}{4}$, respectively
- A machine in fair condition:
  - will remain that way for a mean time $\mu_2$ and
  - will then break down
- A broken machine:
  - will be repaired, which takes a mean time $\mu_3$ and
  - when repaired, will be in good condition and fair condition w.p. $\frac{2}{3}$ and $\frac{1}{3}$, respectively
- Want to find proportions of time for which machine is in each state
- Stationary distribution $\pi_1, \pi_2, \pi_3$ obtained by solving the system of equations:
  - $\pi_1 + \pi_2 + \pi_3 = 1$,
  - $\pi_1 = \frac{2}{3}\pi_3$,
  - $\pi_2 = \frac{3}{4}\pi_1 + \frac{1}{3}\pi_3$,
  - $\pi_3 = \frac{1}{4}\pi_1 + \pi_2$.
- Solution:
  - $\pi_1 = \frac{4}{15}$, $\pi_2 = \frac{1}{3}$, and $\pi_3 = \frac{2}{5}$
- Proportions of time the machine is in different states:
  - $P_1 = \frac{4\mu_1}{4\mu_1+5\mu_2+6\mu_3}$,
  - $P_2 = \frac{5\mu_2}{4\mu_1+5\mu_2+6\mu_3}$,
  - $P_3 = \frac{6\mu_3}{4\mu_1+5\mu_2+6\mu_3}$

# Semi-Markov Processes (contd.)

- Recall that we derived:
  - $P_i = \lim_{t\to\infty} P\{Z(t) = i | Z(0) = k\}$
- We now generalize this result
- Let:
  - $Y(t) =$ time from $t$ until the next transition,
  - $S(t) =$ state entered at the first transition after $t$
- **Theorem**: If the semi-Markov process is irreducible and not lattice, then:

  *1)* $\lim_{t\to\infty} P\{Z(t) = i, Y(t) > x, S(t) = j | Z(0) = k\} = \frac{P_{ij}\int_x^\infty F_{ij}^c(y)dy}{\mu_{ii}}$
- **Proof**:
  - Process can be considered as an alternating renewal process in which:
    - a cycle begins each time the process enters state $i$,
    - process is "on" if the state is $i$ and it will remain $i$ for at least the next $x$ time units and the next state is $j$; otherwise, the process is "off"
  - $E[\text{cycle time}]$:
    - $\mu_{ii}$
  - $E[\text{"on" time in a cycle}]$:
    - $= P_{ij}E\left[\left(X_{ij} - x\right)^+\right]$
    - $= P_{ij}\int_0^\infty P\{X_{ij} - x > a\}da$
    - $= P_{ij}\int_x^\infty F_{ij}^c(y)dy,$
    - where $X_{ij}$ is a r.v. with CDF $F_{ij}$ and representing the time to make a transition from $i$ to $j$, and $y^+ = \max(0, y)$
  - Required limit equals $\frac{E[\text{"on" time in a cycle}]}{E[\text{cycle time}]}$
- **Note**: Long-run proportion of time that the semi-Markov process is in state $i$ and will spend the next $x$ time units without a transition and will then go to state $j$ is also equal to RHS of 1)

# Semi-Markov Processes (contd.)

- Recall:
  - **Theorem**: If the semi-Markov process is irreducible and not lattice, then:

$$1)\quad \lim_{t\to\infty} P\{Z(t)=i, Y(t)>x, S(t)=j \mid Z(0)=k\} = \frac{P_{ij}\int_x^\infty F_{ij}^c(y)dy}{\mu_{ii}}$$

- Using the same technique as used to prove above theorem, we can prove the following result
- **Proposition**: If the semi-Markov process is irreducible and not lattice, then:

$$2)\quad \lim_{t\to\infty} P\{Z(t)=i, Y(t)>x \mid Z(0)=k\} = \frac{\int_x^\infty H_i^c(y)dy}{\mu_{ii}}$$

- We can also prove 2) by summing 1) over $j$
- **Note**: Long-run proportion of time that the semi-Markov process is in state $i$ and will spend the next $x$ time units without a transition is also equal to RHS of 2)

# Branching Processes

- Consider a population consisting of individuals able to produce offspring of the same kind
- Suppose each individual will, by the end of its lifetime, have produced $j$ new offspring w.p. $P_j$, $j \geq 0$:
  - independently of number of offspring produced by any other individual
- Let $X_0$:
  - number of individuals initially present
  - called size of zeroth generation
- All offspring of zeroth generation constitute the first generation
- Let $X_1$:
  - size of first generation
- In general, let $X_n$:
  - size of $n$th generation
- $\{X_n, n \geq 0\}$ is a DTMC:
  - called a *branching process*
- Applications:
  - biology, e.g., spread of bacteria, cell division or mutation
  - physics, e.g., neutrons in a nuclear reactor, cosmic rays
  - epidemiology, e.g., disease spread
  - genealogy, e.g., spread of surnames
  - network analysis, e.g., cascading failures in networks

# Branching Processes (contd.)

- Recall:
  - each individual will, by the end of its lifetime, have produced $j$ new offspring w.p. $P_j, j \geq 0$
  - $X_n$: size of $n$th generation
  - $\{X_n, n \geq 0\}$ is a DTMC
- Is state $0$ positive recurrent, null recurrent, or transient?
  - positive recurrent since $P_{00} = 1$
- **Proposition**: If $P_0 > 0$, then all states other than $0$ are transient
- **Proof**:
  - $P_{i0} = P_0^i > 0$
  - So, starting with $i$ individuals, there is a positive probability of at least $P_0^i$ that no later generation will ever consist of $i$ individuals
- So, w.p. $1$, any finite set of transient states $\{1,2, \dots, n\}$ will be visited only finitely often
- Hence, if $P_0 > 0$, then the population will either die out or its size will converge to infinity

# Branching Processes (contd.)

- Suppose $X_0 = 1$
- Want to find mean and variance of $X_n$
- Let:
  - $Z_i$: number of offspring of $i$th individual of the $(n-1)$st generation
  - $\mu$: mean number of offspring of a single individual, i.e., $\mu = E[Z_i]$
  - $\sigma^2$: variance of number of offspring of a single individual, i.e., $\sigma^2 = \mathrm{var}(Z_i)$
- $X_n$ in terms of $X_{n-1}$ and $Z_1, Z_2, \ldots$:
  - $\sum_{i=1}^{X_{n-1}} Z_i$
- So, $E[X_n]$:
  - $\mu E[X_{n-1}]$
  - $\mu^2 E[X_{n-2}]$
  - $\vdots$
  - $\mu^n$

# Branching Processes (contd.)

- Recall:
  - ❑ $X_0 = 1$
  - ❑ $Z_i$: number of offspring of $i$th individual of the $(n-1)$st generation; $E[Z_i] = \mu$ and $\mathrm{var}(Z_i) = \sigma^2$
  - ❑ $X_n = \sum_{i=1}^{X_{n-1}} Z_i$
  - ❑ $E[X_n] = \mu^n$
- Conditional variance formula:
  1) $\mathrm{var}(X_n) = E[\mathrm{var}(X_n|X_{n-1})] + \mathrm{var}\big(E(X_n|X_{n-1})\big)$
- $E(X_n|X_{n-1})$:
  *2)* $\mu X_{n-1}$
- $\mathrm{var}(X_n|X_{n-1})$:
  *3)* $X_{n-1}\sigma^2$
- By 1), 2), and 3), $\mathrm{var}(X_n)$:
  - ❑ $\sigma^2\mu^{n-1} + \mu^2\mathrm{var}(X_{n-1})$
  - ❑ $\sigma^2\mu^{n-1} + \mu^2\big(\sigma^2\mu^{n-2} + \mu^2\mathrm{var}(X_{n-2})\big)$
  - ❑ $\sigma^2(\mu^{n-1} + \mu^n) + \mu^4\mathrm{var}(X_{n-2})$
  - ❑ $\sigma^2(\mu^{n-1} + \mu^n) + \mu^4\big(\sigma^2\mu^{n-3} + \mu^2\mathrm{var}(X_{n-3})\big)$
  - ❑ $\sigma^2(\mu^{n-1} + \mu^n + \mu^{n+1}) + \mu^6\mathrm{var}(X_{n-3})$
  - ❑ :
  - ❑ $\sigma^2(\mu^{n-1} + \mu^n + \cdots + \mu^{2n-2}) + \mu^{2n}\mathrm{var}(X_0)$
- Hence:
  - ❑ $\mathrm{var}(X_n) = \begin{cases} \sigma^2\mu^{n-1}\left(\frac{1-\mu^n}{1-\mu}\right), & \text{if } \mu \neq 1, \\ n\sigma^2, & \text{if } \mu = 1. \end{cases}$

# Branching Processes (contd.)

- Let $\pi_0$ denote the probability that the population will eventually die out (under the assumption that $X_0 = 1$)
- That is:
  - $\pi_0 = \lim_{n \to \infty} P\{X_n = 0 | X_0 = 1\}$
- **Proposition**: $\pi_0 = 1$ if $\mu < 1$
- **Proof**:
  - By Markov's inequality:
    - $P\{X_n \geq 1\} \leq E(X_n) = \mu^n \to 0$ as $n \to \infty$
  - So, $P\{X_n = 0\} \to 1$ as $n \to \infty$
- Later, we will show that $\pi_0 = 1$ even when $\mu = 1$

# Branching Processes (contd.)

- Recall:
  - $\pi_0$ denotes the probability that the population will eventually die out (under the assumption that $X_0 = 1$)
- **Proposition**: $\pi_0$ satisfies the following equation:
  - $\pi_0 = \sum_{j=0}^{\infty} \pi_0^j P_j$
- **Proof**:
  - Assume that $X_0 = 1$
  - $\pi_0$:
    - $P\{\text{population dies out}\}$
    - $\sum_{j=0}^{\infty} P\{\text{population dies out}|X_1 = j\} P_j$
    - $\sum_{j=0}^{\infty} \pi_0^j P_j$, since if $X_1 = j$, then each of the $j$ families independently dies out w.p. $\pi_0$

# Branching Processes (contd.)

- Assume that $X_0 = 1$
- **Theorem**: Suppose $P_0 > 0$ and $P_0 + P_1 < 1$. Then:
  - *a)* $\pi_0$ is the smallest positive number satisfying
    - $\pi_0 = \sum_{j=0}^{\infty} \pi_0^j P_j$
  - *b)* $\pi_0 = 1$ iff $\mu \leq 1$
- E.g. 1:
  - Suppose $P_0 = \frac{1}{2}$, $P_1 = \frac{1}{4}$, $P_2 = \frac{1}{4}$
  - $\mu$:
    - $\frac{3}{4}$
  - $\pi_0$:
    - $1$
- E.g. 2:
  - Suppose $P_0 = \frac{1}{4}$, $P_1 = \frac{1}{4}$, $P_2 = \frac{1}{2}$
  - $\mu$:
    - $\frac{5}{4}$
  - $\pi_0$ satisfies the equation:
    - $\pi_0 = \frac{1}{4} + \frac{1}{4}\pi_0 + \frac{1}{2}\pi_0^2$
    - i.e., $2\pi_0^2 - 3\pi_0 + 1 = 0$
  - Smallest positive solution:
    - $\pi_0 = \frac{1}{2}$
- In the above examples, what is the probability that the population will die out if it initially consists of $n$ individuals?
  - $\pi_0^n = \begin{cases} 1, & \text{in E.g. 1,} \\ \left(\frac{1}{2}\right)^n, & \text{in E.g. 2.} \end{cases}$

# Branching Processes (contd.)

- Recall:
  - We assume that $X_0 = 1$
  - **Theorem**: Suppose $P_0 > 0$ and $P_0 + P_1 < 1$. Then:
    - *a)* $\pi_0$ is the smallest positive number satisfying
      - *1)* $\pi_0 = \sum_{j=0}^{\infty} \pi_0^j P_j$
    - *b)* $\pi_0 = 1$ iff $\mu \leq 1$
- **Proof**:
  - We omit the proof of part (b) and prove part (a)
  - Suppose $\tilde{\pi}$ satisfies 1)
  - Using induction, we will show that:
    - *2)* $\tilde{\pi} \geq P\{X_n = 0\}, \forall n$
  - Then, since $\pi_0 = \lim_{n\to\infty} P\{X_n = 0\}$, part (a) will follow by taking limits as $n \to \infty$ on both sides of 2)
  - $\tilde{\pi} = \sum_{j=0}^{\infty} \tilde{\pi}^j P_j \geq \tilde{\pi}^0 P_0 = P_0 = P\{X_1 = 0\}$
  - Assume that $\tilde{\pi} \geq P\{X_n = 0\}$
  - Then $P\{X_{n+1} = 0\}$:
    - $= \sum_j P\{X_{n+1} = 0 | X_1 = \mathrm{j}\} P_j$
    - $= \sum_j (P\{X_n = 0\})^j P_j$
    - $\leq \sum_j \tilde{\pi}^j P_j$ (by the induction hypothesis)
    - $= \tilde{\pi}$
  - This completes the proof of part (a)